App for Navigating Ecological Surveys

CS310 Kailo Bouskila March 2023

Ecological Surveys

- Collecting data on populations of organisms in an ecosystem.
- Sampling a **subset** of the population rather than the entire population.
- Used to **estimate** the abundance, density or distribution of the **whole** population.
- Can identify trends and patterns over time.
- There exist many methods of sampling

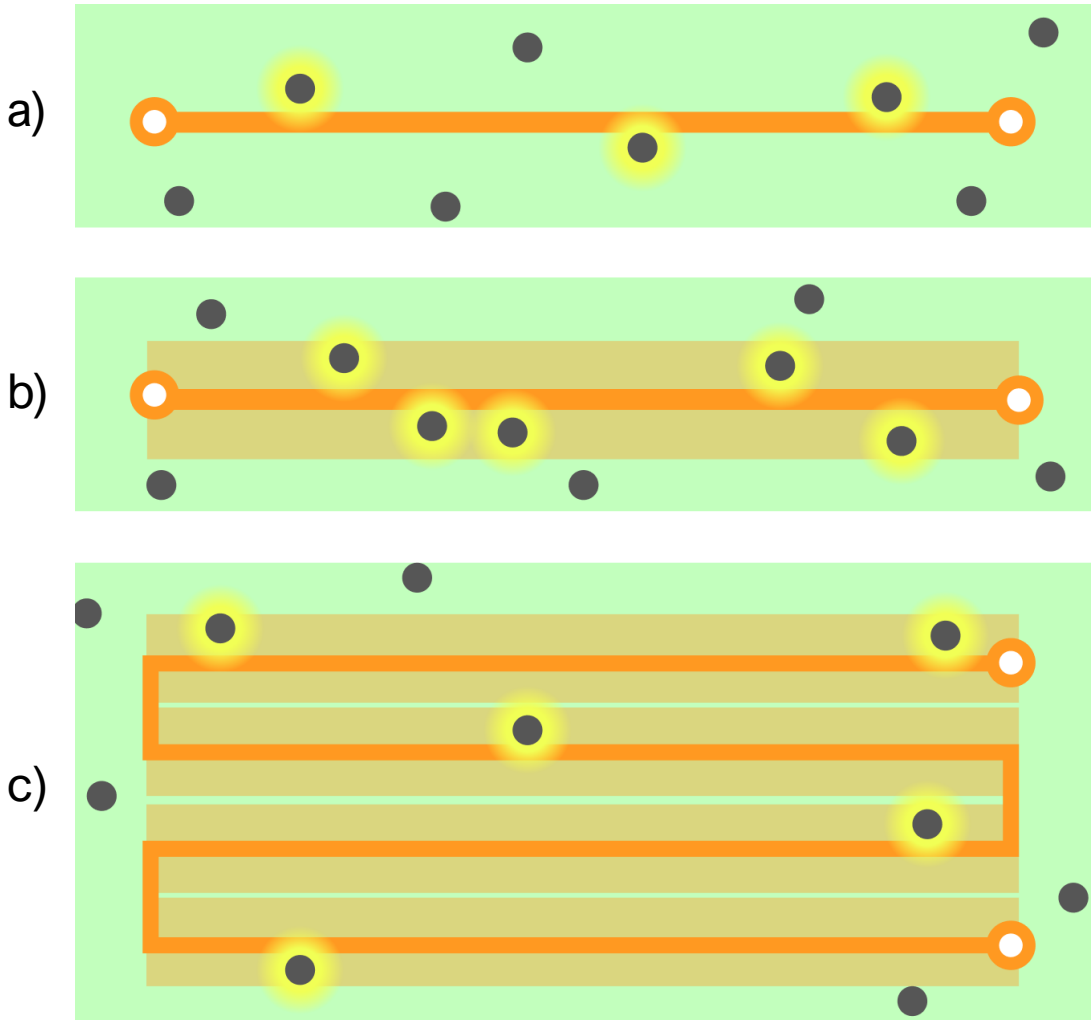
Transect Sampling

- Recording the presence of objects along a line or path.

  a) Line transects – recording data on object along a line

  b) Strip transects – recording data on objects inside wider strip

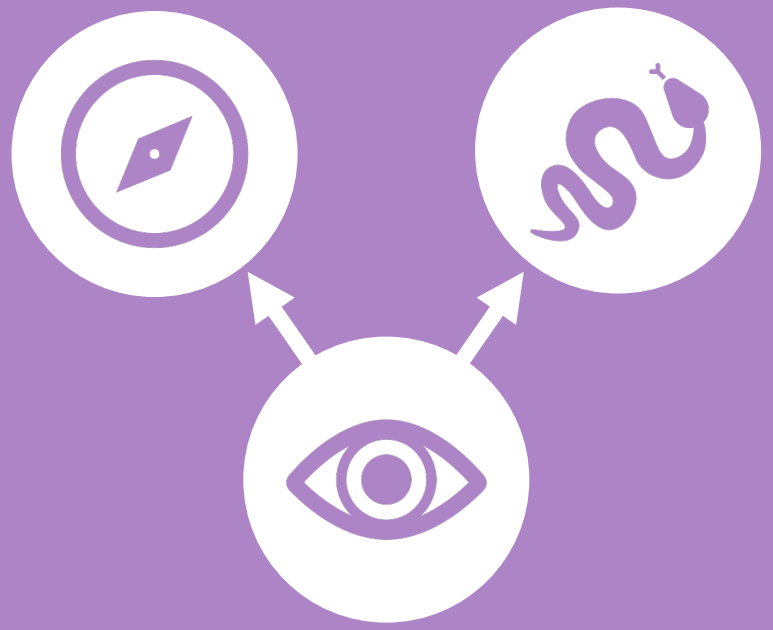
  c) Multiple strip transects



The problem

The surveyor must be:

- Attentive
  - Animals can be evasive or hard to spot
- On track
  - Observations must be from the set path
- Focused
  - To keep track of observations



Existing Technologies

Sensor Networks

- Used in ecology to measure distribution of air/soil/water quality across an area.
- Can involve motion-activated cameras to detect animals
- Collect data automatically and over long periods of time.
- Special equipment required
- Long setup time

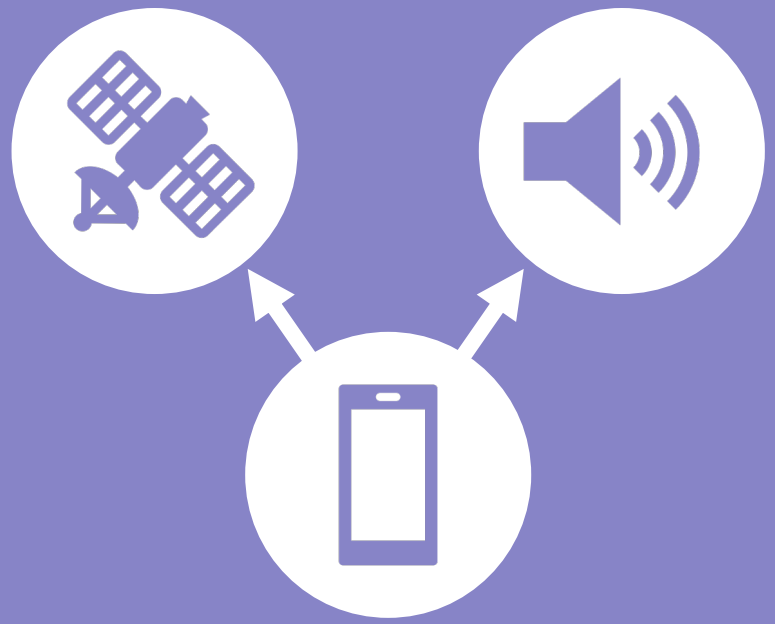
Arial Mapping

- Drones can fly along a pre-programmed flight path, stitching together many photos.
- Produces a detailed aerial image of the survey site.
- Can’t see past dense trees or foliage
- Good for large organisms only

The solution

Mobile phones:

- Can multitask
  - Can run in the background or in your pocket
- Are GPS enabled
  - Works globally without internet
- Can navigate using audio cues
  - Audio instructions can navigate a user



Existing Applications

Google Maps

Audio navigation

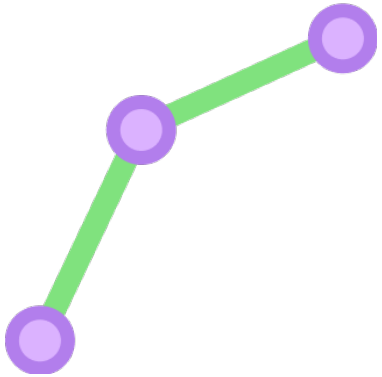
AllTrails

Trail navigation

Survey Types

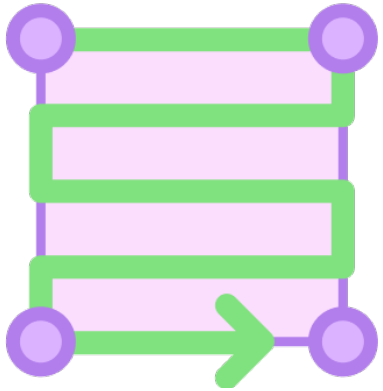
Line transect

One or more connected points.



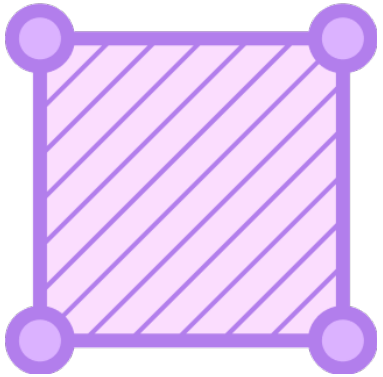
Multiple Strip Transects

Area subdivided into equal-width strips.

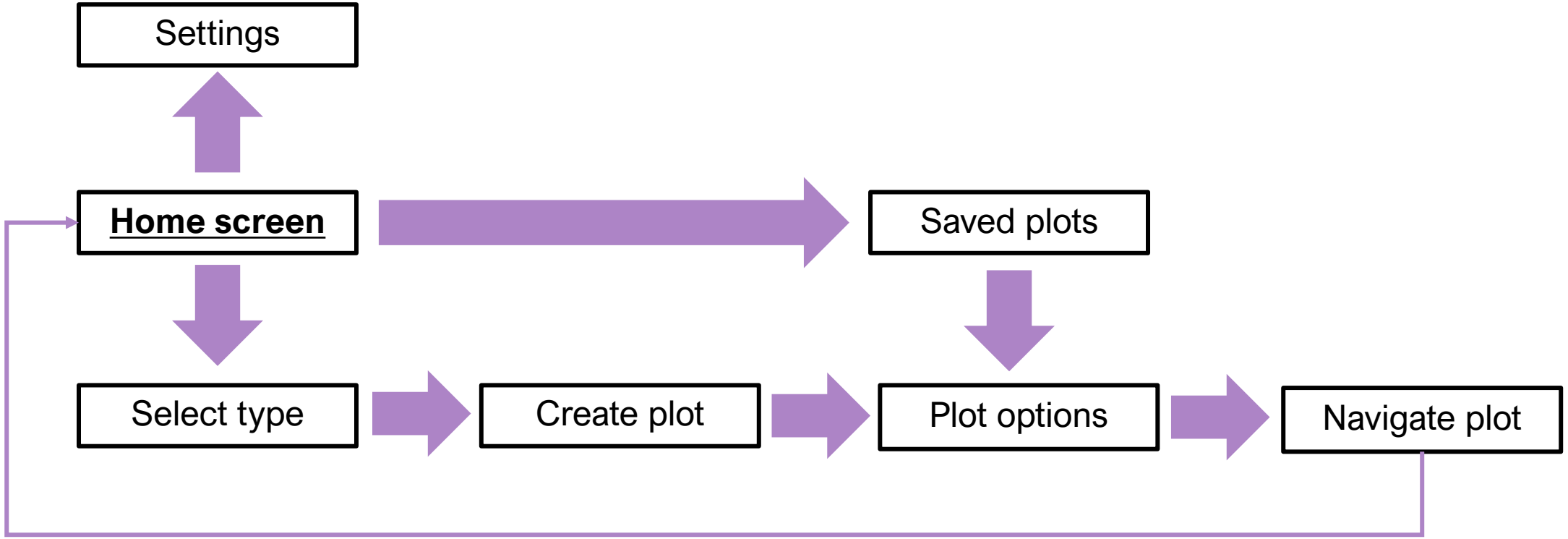


Area plot

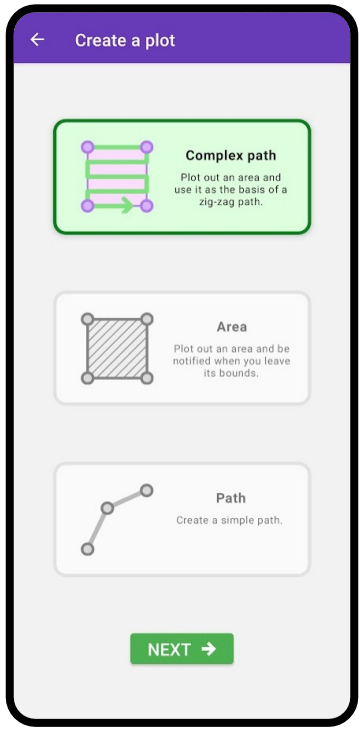
Notifies user when they leave the area



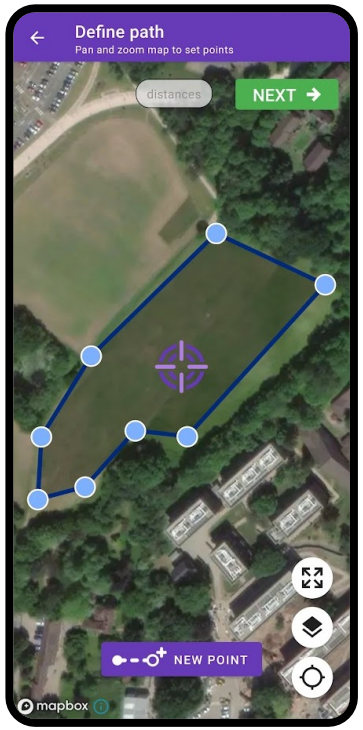
App Outline


Settings
Home screen
Saved plots
Select type
Create plot
Plot options
Navigate plot

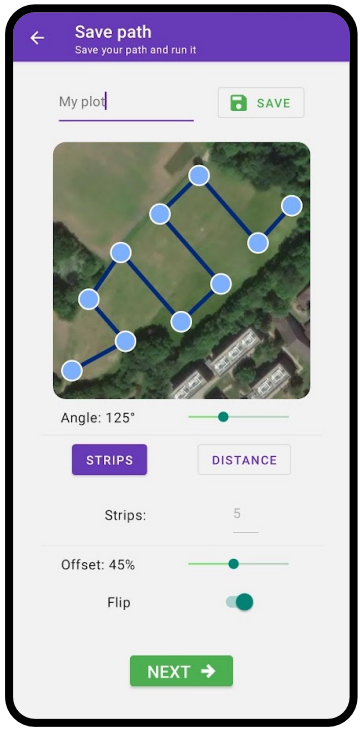
User Interface



Select type



Create plot



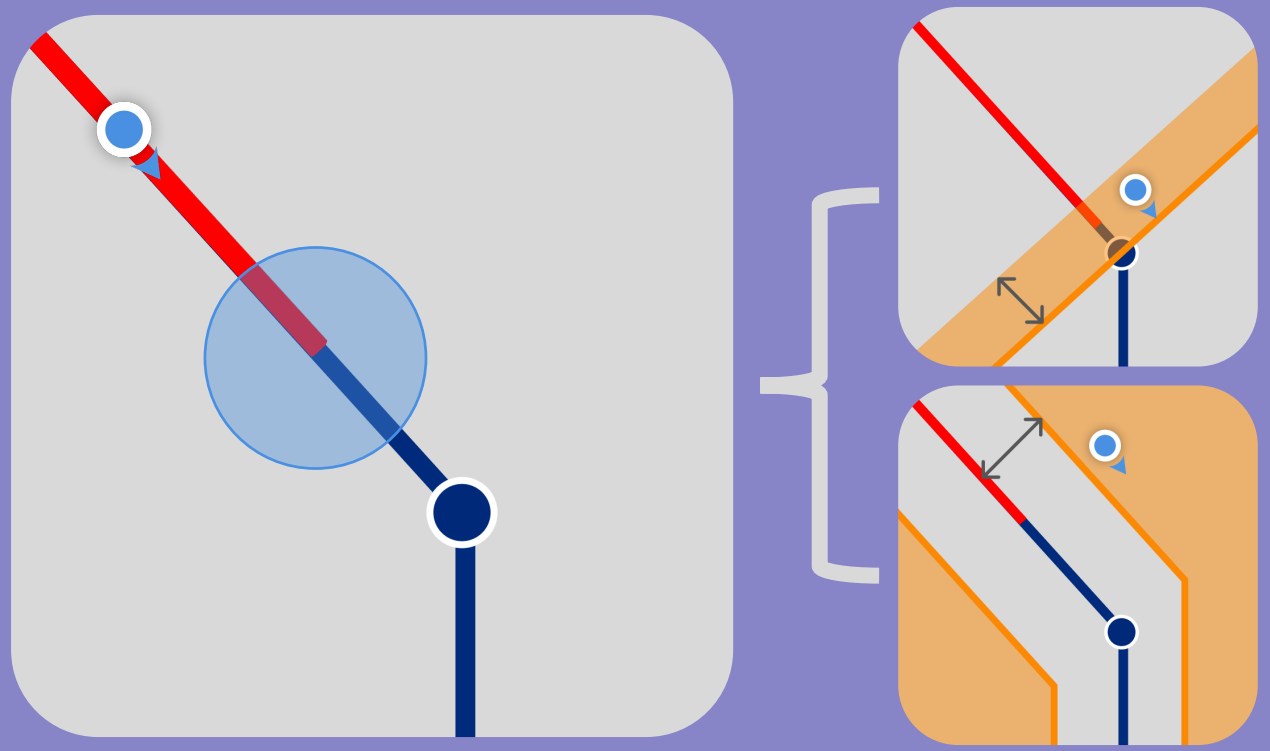
Plot options

Navigate plot

Development

- React-Native - Framework for JavaScript
- Expo framework – Framework for React-Native
- Mapbox API

Navigation Algorithm



1) User is in range of the end of the segment

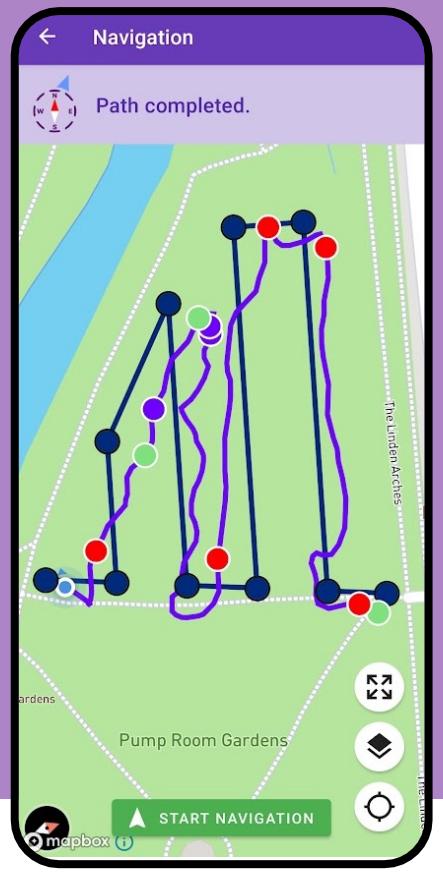
   instruct them to the next point

2) User has strayed off course

   instruct them to keep left/right

Testing

Foreground



Background

App was most accurate when moving slowly, battery saver off, app running in foreground.

Transect path

Recorded location

Points where instructions were read

Direction to next point

User is off course

User is back on track

Evaluation & Limitations

- Main limitation is **mobile GPS accuracy** (~5m)
  - Depends on weather conditions, device, battery life, foreground/background.
- Works better on large survey plots than small ones.
- Instructions can sometimes be read at the wrong time
  - Navigation sensitivity needs to be set carefully
- Drains battery life

Future Work

Offline capability

Maps could be downloaded offline

Sharing plots between users

Using a shared file or QR code

Additional tools for recording data

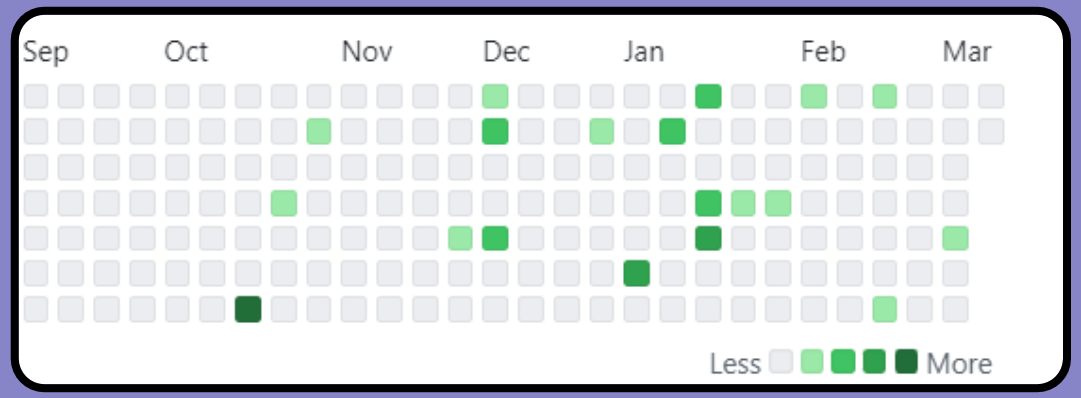
Not just aiding with navigation but with object observations as well

Additional surveying methods

Such as belt transects, distance sampling and curved line transects

Project Management

- Development was mostly smooth and on schedule.
- Combination of agile and plan-based approach – new iteration every 2-3 weeks
- Supervisor meeting every week

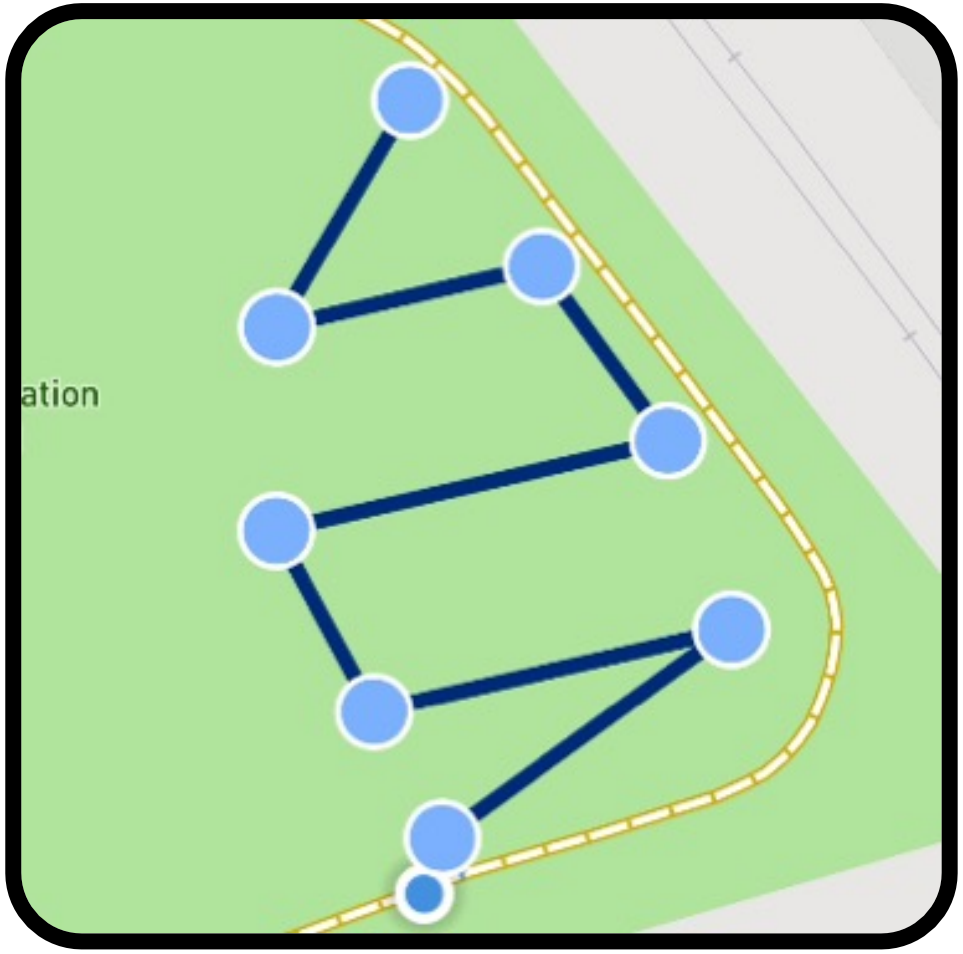


GitHub commit history

Unforeseen problems:

- Library compatibility issues
- Delays in making the app work in the background

Demo



References

- D. Hill, M. Fasham, G. Tucker, M. Shewry, and P. Shaw, Eds., *Handbook of Biodiversity Methods: Survey, Evaluation and Monitoring*. Cambridge: Cambridge University Press, 2005
- Rundel PW, Graham EA, Allen MF, Fisher JC, Harmon TC. *Environmental sensor networks in ecological research.* The New Phytologist, 182(3), 589–607, 2009